



Research paper

24 loci MIRU-VNTR analysis and pattern of drug resistance in pre-extensively drug resistant pulmonary tuberculosis in Bangladesh

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ABSTRACT

Phylogenetic diversity and distinct phylogeographic distribution of *Mycobacterium tuberculosis* (MTB) contribute to regional differences in drug resistance. The emergence of pre-extensively drug resistant tuberculosis (Pre-XDR-TB) becomes obstacles to achieve End TB strategy in Bangladesh. This cross-sectional study was conducted to identify the strains of different lineages of MTB, their variations of distribution among Pre-XDR-TB cases and to observe the linkage of particular strains of MTB with drug resistance. A total of 33 Pre-XDR-TB isolates were enrolled in this study. All isolates were confirmed as MTB by MPT 64 antigen detection and genotyped by 24 loci Mycobacterial Interspersed Repetitive Unit-Variable Number of Tandem Repeats (MIRU-VNTR) analysis. Drug resistance was detected by second line Line probe assay (LPA). Beijing was the predominant strain 16 (48.48%), followed by Delhi/CAS 5 (15.15%), LAM 4 (12.12%) and Harlem 3 (9.10%), EAI 2 (6.06%), Cameroon 2 (6.06%) and NEW-1 1 (3.03%). There were 31 different genotypes consisting of 2 clusters and 29 singletons. All the clustered strains were belonged to Beijing lineage. Recent transmission occurred mainly by Beijing strains, showed low transmission rate (12.1%). Of 33 isolates 30 (90.90%) were Fluoroquinolones resistant, the mutations involved was Asp94Gly in gyr A MUT 3C gene 13 (39.39%) in quinolone resistance determining region (QRDR) followed by 11 (33.33%) in gyr A MUT 1. Three (9.10%) isolates showed resistant to injectable 2nd line drugs and all mutation occurs in G1484T of rrs MUT 2. Beijing lineage was predominant in treatment failure and relapse cases. Levofloxacin was resistant to all Pre-XDR-TB cases, but moxifloxacin showed low level resistance. QUB 26 was the most discriminatory locus (0.85) among 24 loci whereas MIRU 2 was the least (0.03). 24 loci MIRU-VNTR analysis shows high discriminatory index (0.71), found to be powerful tool for genotyping of Pre-XDR-TB, which is the first study in Bangladesh that enhanced the current TB control policy.

1. Introduction

Tuberculosis (TB) is a major public health problem in Bangladesh. Worldwide, TB is the 13th leading cause of death (WHO, 2019. Global Tuberculosis Report 2019. World Health Organization, Geneva). It is the second leading infectious killer after COVID-19 (above HIV/AIDS). In 2020, 86% of new TB cases occurred in the 30 high TB burden countries. Eight countries accounted for two third of the new TB cases: India, China, Indonesia, The Philippines, Pakistan, Nigeria, Bangladesh and South Africa (WHO, 14 October, 2021). Bangladesh is among 30 high TB burden and MDR/RR-TB burden countries.

End TB Strategy of World Health Organization is 95% reduction of TB deaths and 90% reduction in TB incidence rate by 2035. Globally, the

TB mortality rate is falling at about 3% per year. TB incidence is falling at about 2% per year and between 2015 and 2020 the cumulative reduction was 11%. This was over halfway to the End TB Strategy milestone of 20% reduction between 2015 and 2020 (WHO, 14 October, 2021).

We have entered in the era of End TB strategy from Stop TB strategy. MDR-TB is one of the major problems regarding achieving the goals. Recently Pre-extensively drug resistant (Pre-XDR) and extensively drug resistant (XDR) TB have become areas of growing concern.

The prevalence of Pre-XDR-TB differs across the world due to many factors. In addition of the environmental and host factors, the genetic diversity of *Mycobacterium tuberculosis* Complex (MTBC) may play a role because it shows distinct phylogeographic distribution (Coscolla and

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Gagneux, 2014).

Phylogenetic diversity and distinct geographic distribution of *M. tuberculosis* (MTB) plays major role to regional differences in drug resistance. The emergence of pre-extensively drug resistant tuberculosis (Pre-XDR-TB) becomes obstacles to achieve End TB strategy in Bangladesh. Pre-XDR-TB is defined as TB with resistance to isoniazid and rifampicin and either a FQ or a second-line injectable agent but not both (Singhal et al., 2016). In Bangladesh prevalence of Pre-XDR-TB was (16.18%) (Tasnim et al., 2018). In India prevalence of Pre-XDR-TB were 55.65% among MDR-TB patients studied in a tertiary care hospital of Mumbai and 49.4% in north India (Adwani et al., 2016; Singhal et al., 2016). A South African study showed 14% of multidrug-resistant isolates were found to be Pre-XDR TB (Banerjee et al., 2008). Molecular epidemiological studies of *M. tuberculosis* identified variability in the phylogeography of strains globally (Coscolla and Gagneux, 2014).

The commonly found strains worldwide are Beijing, Cameroon, Delhi/Central Asian (CAS), East African Indian (EAI), Ghana, Haarlem, Latin American-Mediterranean (LAM), Uganda I and II, Ural, and X lineages (Allix-Béguec et al., 2008). The types of strains vary country to country as well as regionally. Beijing strains are most prevalent globally and also associated with enhanced acquisition of drug resistance; however, their resistance patterns varied regionally (Eldholm et al., 2015).

To detect strains of *M. tuberculosis* several typing methods are used which includes IS6110 based RFLP, Single nucleotide polymorphisms (SNPs), Deletion analysis, MIRU-VNTR, Spoligotyping Whole genome sequencing (WGS) technologies are usually used. MIRU-VNTR typing has become an important method as it allows high throughput, discriminatory and reproducible analysis of clinical isolates (Jagielski et al., 2014; Desikan and Narayanan, 2015). 24-loci Mycobacteria interspersed repetitive units-variable number of tandem repeat (MIRU-VNTR) can provide unique high-resolution insights into the population structure of the *M. tuberculosis* and provides clear criteria for the identification of the different *M. tuberculosis* lineages and sub-lineages (Allix-Béguec et al., 2008). Now MIRU-VNTR becomes newly recommended gold standard for phylogenetic analysis of *M. tuberculosis* (Jagielski et al., 2014).

WHO recommended the use of molecular line probe assay (LPA) for rapid screening of MDR-TB, Pre-XDR-TB and XDR-TB in low- and middle-income settings. LPA use multiplex PCR amplification and hybridization to identify *M. tuberculosis* complex and mutation to genes associated with different drug resistance. Second Line LPA shows good specificity and sensitivity for the detection of FLQ resistance ($\geq 86\%$ and $\geq 99\%$) and second line injectable drug (SLID) resistance ($\geq 87\%$ and $\geq 99\%$) compared to phenotypic drug sensitivity testing on culture isolate or directly sputum specimen (WHO, 2016). Although the prevalence of Pre-XDR-TB alarmingly increases in Bangladesh, till now to the best of our knowledge, no published data on mutations and genotypes of Pre-XDR-TB strains are currently available in Bangladesh.

To gain an insight about molecular typing of *M. tuberculosis*, 24 loci MIRU-VNTR was performed in this study. Drug resistance-associated mutations in Pre-XDR-TB isolates from Bangladesh was also documented using 2nd line LPA (MTBDRsl).

2. Materials and methods

2.1. Study place

This cross-sectional study was conducted from March 2018 to February 2019 at Department of Microbiology & immunology, Bangabandhu Sheikh Mujib Medical University (BSMMU) and National Tuberculosis Reference Laboratory (NTRL), Dhaka, Bangladesh.

2.2. Sample collection

A total of 98 pulmonary MDR-TB isolates diagnosed by Xpert MTB/RIF assay were collected during study period. Of which 33(33.67%)

cases were identified as Pre-XDR-TB. Microscopy, DNA extraction and second line LPA was done in NTRL, 24 loci MIRU-VNTR typing of 33 Pre-XDR-TB isolates were performed at Department of Microbiology & Immunology, BSMMU. The confirmed Multi drug resistant (MDR) MTB sample isolates were collected at NTRL. These isolates were cultured on both Lowenstein-Jensen (L-J) media and in Mycobacterial Growth Indicator Tubes (MGIT). Isolates were identified as *M. tuberculosis* by MPT 64 antigen test (Standard Diagnostic Inc., Korea). Phenotypic resistance of both first and second line of drugs was performed in MGIT 960 system. DNA extraction was performed following standard operating procedure (Kumar et al., 2011). Extracted DNA was transferred to BSMMU in screw cap vials within cold box to perform conventional PCR for MIRU-VNTR analysis. *M. tuberculosis* H37Rv (ATCC 27294) was used as reference strain in this study.

2.3. 24 loci MIRU-VNTR typing and analysis

MIRU-VNTR typing is based on amplification by PCR. It can be performed on heat inactivated mycobacterial colonies or mycobacterial pellets from liquid cultures without extensive DNA purification (Supply et al., 2006). Qiagen Hotstart Taq Master Mix kit including Q solution (Qiagen, Germany) and primers (Imperial life science, India) were designed as sequence found at Supply P, 2005 (Supply, 2005) were used for amplification. Amplification was carried out in an automated DNA thermal cycler (Applied bio-system 2720). PCR cycle run as follows: 15 min at 95 °C, 1 min at 94 °C, 30 s at 59 °C, 1 min 30 s at 72 °C (these were repeated for 40 cycles), 10 min at 72 °C, and store at 4 °C. The amplified product was detected by Gel electrophoresis was done in 3% agarose for 5 h at 120 constant Voltage. Gel was stained with ethidium bromide to detect the specific DNA bands. Marker of DNA of known bp (100 bp ladder) was used to determine the size of amplified PCR products. The DNA bands of the samples were visualized by using a UV transilluminator and photograph was taken for further documentation.

2.4. Strain identification

There was a standard allelic table which was found from the database website. The number of repeats for each locus was determined based on this allelic table. Ultimately a numerical data was found for 24 loci for every isolate based on this allelic table. The numerical data was assorted on a predesigned Excel spread sheet and exported to the reference data base. MIRU-VNTR plus web tool (<http://www.MIRU-VNTRplus.org>) was used for identification and comparison of strains by similarity search and phylogenetic tree-based analysis (Neighbour-Joining).

2.5. Molecular drug susceptibility testing by line probe assay (Laboratory manual mycobacteriology laboratories, TB alliance, 2014)

Genotype MTBDRsl assay was performed according to the instructions provided by the manufacturer (Hain Life science GmbH, Nehren, Germany, n.d) for second line LPA (Hain Life Sciences). The extracted DNA was amplified by using HAIN amplification master mix. Amplification at PCR and Hybridization of the amplified sample were performed according to the instructions. There were both mutation probes and wild type probes for each gene. The presence of any mutation band regarded as resistance to the respective antibiotic. Absence of at least one of the wild types was also regarded as resistant.

2.6. Data analysis

The number of repeats for each locus was determined based on the allelic table by Supply et al. (2006) and clustering analysis was done using the online tool at <http://www.MIRU-VNTRplus.org>. Statistical analysis of data was done with SPSS software (ver.23). Descriptive analysis of all relevant variables was done by using frequency, percentage, mean, table and figure etc. Hunter Gaston discriminatory index

(available at http://insilico.ehu/mini_tools/discriminatory_power/) was calculated to observe discriminatory power of 24 loci MIRU-VNTR (Hunter and Gaston, 1988).

2.7. Ethical approval

The study was approved by the Ethics committee of Institutional Review Board of Bangabandhu Sheikh Mujib Medical University (BSMMU), Dhaka, Bangladesh (Approval Number: BSMMU/2018/8047, July 16, 2018). Written approval was taken from the coordinator of NTRL to collect confirmed multi drug resistant (MDR) *M. tuberculosis* isolates for this study purpose. Informed written consent was taken from the patient. Data were anonymized to maintain the confidentiality and were used for research purpose only.

3. Results

3.1. 24 locus MIRU-VNTR typing

Among 33 Pre-XDR-TB isolates 24 locus MIRU-VNTR typing revealed Beijing is the predominant strain 16 (48.48%), followed by Delhi/CAS 5 (15.15%), LAM 4 (12.12%) and EAI 3 (9.10%). Other strains were Harlem 2 (6.06%) Cameroon 2 (6.06%) and NEW-1 one (3.03%) (Table 1).

Minimum spanning tree analysis through MIRU-VNTR plus web tool showed 4 clonal complexes. Largest clonal complex is formed by 5 Beijing strains. Another 3 clonal complex formed by Beijing and Delhi/CAS strains consisting of 2 strains of each lineage. Rest of the 22 isolates was remaining as Singletons. Four strains arranged in 2 cluster and these clustered isolates belongs to Beijing lineage (Fig. 1).

3.2. Socio demographic data and risk factors associated with pre-XDR-TB isolates

Among 33 Pre-XDR-TB isolates 24 cases were male (72.72%) that were significantly higher than the 9 female (27.28%) patients. Most patients were within 30–39 years of age group 9 (27.27%). Urban populations were greater in number 27 (81.81%) than the rural populations. Most of the patients were self-employed 12 (36.36%) by profession. Almost all are BCG vaccinated 29 (87.87%). Most of the patient having no history of drug and alcohol misuse Population characteristics among the Lineages also indicate 12 (50%) cases out of 24 of the previously treated patients with anti-TB drugs were belongs to Beijing family (Table 2, supplementary Table 3).

3.3. Drug resistance pattern of Pre-XDR-TB isolates

Supplementary Table 2 showed Drug Sensitivity pattern of 33 Pre-XDR-TB isolates. Among them 30 (90.90%) were Fluoroquinolones resistant. Only 3 (9.10%) isolates showed resistant to injectable 2nd line drugs. 2nd line LPA determines the resistance pattern of 33 Pre-XDR-TB isolates 13 (39.39%) showed point mutation at MUT 3C {gyr A D94G (GAC-GGC)} followed by 11 (33.33%) Pre-XDR-TB isolates those showed point mutation at MUT 1 {gyr A A90V (GCG-GTG)}. Another 3

(9.10%) Pre-XDR-TB isolates showed point mutation at MUT 3B {gyr A D94N (GAC-AAC) D94Y (GAC-TAC)}. Only one (3.03%) Pre-XDR-TB isolates showed point mutation at MUT 2 {gyr A S91P (TCG-CCG)}. Furthermore 2 (6.06%) of the Pre-XDR-TB isolates showed point mutation at MUT 2 {gyr B E540V (GAA-GTA)}. Rest of the 3 (9.10%) Pre-XDR-TB isolates showed resistant to injectable 2nd line drugs. They showed point mutation at MUT 2 {rrs 1484(G-T)} corresponding rrs WT2 band also developed in these cases, that regarded as hetero resistant strain or presence of more than one *M. tuberculosis* complex strain in specimen (Table 3).

Correlation of Fluoroquinolones and injectable 2nd line drug resistance with MTB lineage among Pre-XDR-TB isolates has been shown on Table 4. Beijing Lineage 16 (48.48%) showed maximum mutation in MUT 3C 7 (21.21%). Beijing lineage 2 (6.06%) also showed mutation in gyr B MUT 2. Cameroon lineage showed resistance only in gyr A MUT1 2 (18.18%). Delhi/CAS lineage 3 (9.10%) showed resistance in gyr A MUT 1, one (3.03%) in gyr A MUT 3B. LAM lineage 4 (12.12%) showed mutation only in gyr A, 3 (9.10%) showed mutation in MUT 3C and only one (3.03%) showed mutation in MUT 1. EAI showed mutation in gyr A MUT 3C one (3.03%) and mutation in gyrA 3B 2 (6.06%) Harlem showed mutation only in gyr A among them one (3.03%) in MUT 2 and one (3.03%) in MUT 3C. only one NEW-1 case show mutation in MUT3C 1 (3.03%). Only 3 (9.10%) Pre-XDR-TB isolates showed resistant to injectable 2nd line drugs. They showed point mutation at MUT 2 {rrs 1484(G-T)} corresponding rrs WT2 band also developed in these cases, that regarded as heteroresistant strain. Out of these 3 strains one belongs to Delhi/CAS and rest of 2 strains belongs to Beijing family.

3.4. Association of Pre-XDR-TB isolates with TB treatment outcome

Newly diagnosed Pre-XDR-TB cases are 10 (30.30%) in number among them Beijing was predominant (15.15%). There were 8 (24.24%) Cat 2 failure cases and 7 (21.21%) Cat 2 relapse cases those turned into Pre-XDR-TB, Beijing (12.12%) was also predominant strains among Cat 2 failure and relapse cases. Four cases (12.12%) of Cat 1 failure and 3 cases (9.10%) of Cat1 relapse also turned into Pre-XDR-TB. Only one case (3.03%) of Cat 1 non converter turned into Pre-XDR-TB and that was an EAI strain (Table 5).

3.5. Allelic diversity of 24 MIRU-VNTR loci

Allelic diversity among 24 MIRU-VNTR loci shows highest (0.85) discriminatory locus was 4052/QUB 26. Other very discriminatory loci (>0.6) were Mtub04, 960/MIRU10, 1644/MIRU16, 1955/Mtub21, 2163b/QUB11b, 2165/ETRA, 2996/MIRU26, 3192/MIRU31, 3690/Mtub39, 4156/QUB4156 (>0.6). The lowest (0.03) discriminatory locus in this study was 154/MIRU02 and 2687/MIRU24. Allelic diversity was calculated according to Hunter-Gaston discriminatory index (Supplementary Table 4).

4. Discussion

Drug-resistant tuberculosis continues to be a public health crisis. Tuberculosis continues to remain one of the major health problems in the South-East Asia Region of WHO. MDR-TB is one of the major problems regarding achieving the goals of End TB strategy. Recently extensively drug resistant (XDR) and pre extensively drug resistant (Pre-XDR) TB have become areas of growing concern. The development of resistance to anti tubercular drug reduces the efficacy of standard treatment and arise a vicious challenge to achieve the goals of End TB strategy.

Among Pre-XDR-TB strains, Beijing was the predominant in comparison with the other lineages in this study. Other major Pre-XDR-TB strains included Delhi/CAS and LAM, EAI, Harlem, Cameroon, and NEW-1. In an Indian study, 55.8% of the Pre-XDR-TB isolates were of Beijing genotype followed by Delhi/CAS (34.8%), LAM (4.6%) and two were undetermined (Singhal et al., 2016). A study in China among 30

Table 1
Distribution of MTB strains among Pre-XDR-TB isolates (n = 33).

Lineage	Pre-XDR-TB isolates	
	Number	Percentage
Beijing	16	48.48
Delhi/CAS	5	15.15
LAM	4	12.12
Harlem	3	9.10
EAI	2	6.06
Cameroon	2	6.06
NEW 1	1	3.03
Total	33	100

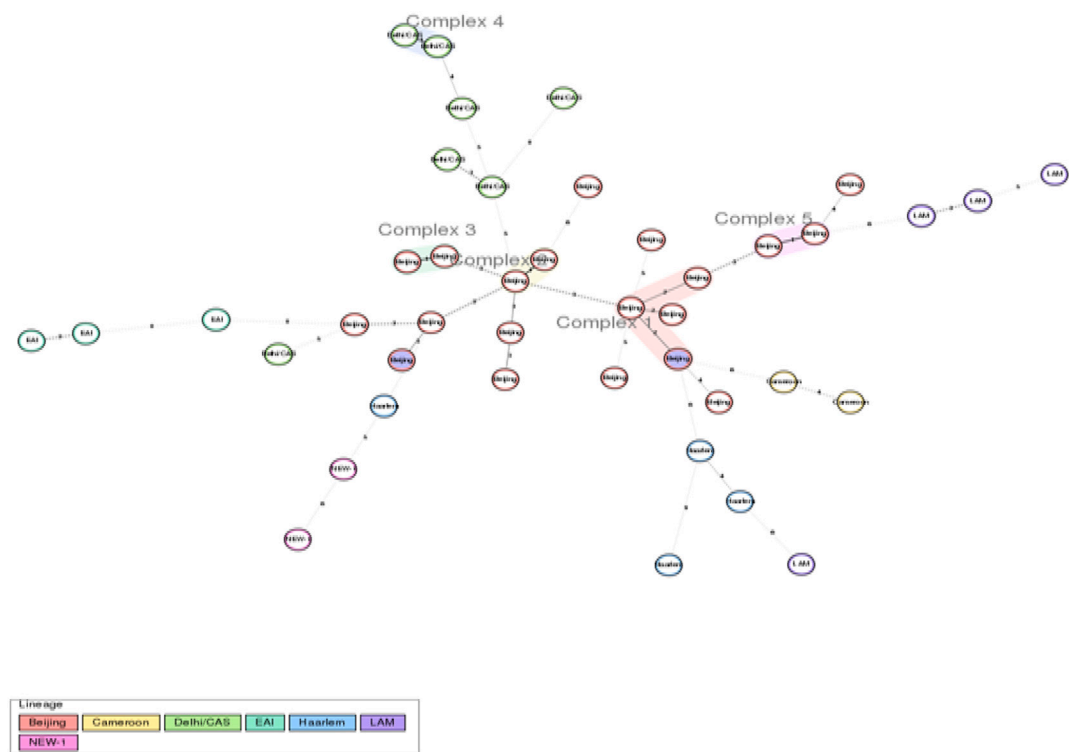


Fig. 1. Minimum spanning tree image of Pre-XDR-TB isolates generated through MIRU-VNTR plus web tool.

Table 2
Socio demographic data of study population (n = 33).

Categories	Sub categories	Pre-XDR (n = 33)
Sex	Male	24 (72.72)
	Female	9(27.28)
Age	15–19	1 (3.03)
	20–29	8 (24.24)
	30–39	9 (27.27)
	40–49	8 (24.24)
	50–59	2 (6.06)
	60–69	6 (18.18)
	70–79	1(3.03)
Location	Urban	27(81.81)
	Rural	6 (18.18)
Occupation	Service	3 (9.10)
	Business	6 (18.18)
	Housewife	7 (21.21)
	Student	5 (15.15)
	Self-employed	12 (36.36)
H/O living outside Bangladesh	Yes	4 (12.12)
	No	29 (87.87)
BCG vaccine mark	Present	29 (87.87)
	Absent	4 (12.12)

pre-XDR-TB isolates revealed 83.1% *M.tuberculosis* isolates belonged to the Beijing family (Hu et al., 2015). Another study in Japan among 38 MDR isolates found Beijing (62%) as the predominant lineage followed by T (13%), LAM (5%), U (2%), EAI (2%) and X (2%) (Murase et al., 2010). Regarding genotyping only one study was conducted among MDR-TB patients of Bangladesh by Banu et al. Most of the strains were belonged to Beijing lineage (19%). About 10% of the strains were EAI. Delhi/CAS, T, LAM family were also identified in that study (Banu et al., 2012a). In another study conducted in a rural area of Bangladesh, strong association was found between the Beijing genotype and drug resistance. They found proportion of MDR-TB was about 24% in the Beijing lineage, compared to about 3% in the non-Beijing lineage (Banu et al., 2012b).

Lineage wise comparison with population characteristics indicates distribution of male predominance in all lineages except LAM, which affects male and female equally, and NEW-1 which affect only male. All age group was infected with diverse group of lineages whereas all patients of 15–19 years of age group and 70–79 years of ages were infected by only Beijing strains. Four (4) patients gave history of living outside of Bangladesh for many years. Two (2) of them were infected by Cameroon strains and 1 of each infected by NEW-1 and Delhi/CAS strains. It is difficult to prove that if they were infected from the outside of Bangladesh or not. Finding of these strains are really matter of suspicion. Patient infected with Delhi/CAS strains lived on Najafgarh in Delhi and patient infected by NEW-1 resides on Germany for many years. NEW-1 previously known as Ural 2 was predominantly found on Ural region of Russia, Kazakhstan, Kirgizstan and Iran. One of the patients infected with Cameroon strains lived in Saudi Arabia and another one lived in Libya for many years. Cameroon is a specialist sub lineage of lineage 4 which showed geographical restrictions. More than 7% of MDR strains were found to be by Cameroon strains in a nationwide study in Saudi Arabia and Libya which is much nearer of its endemic region (Al-Hajjaj and Varghese, 2015; Stucki et al., 2016). Beijing family is the predominant lineage in East and South-East and Eastern Europe. Several studies found strong.

link between Beijing lineage and drug resistance (Borrell and Gagneux, 2009). A study in Myanmar also found that Beijing genotype was significantly higher than EAI among MDR isolates (Phyu et al., 2009). Another finding is that Beijing strains are more common where BCG vaccination coverage is more. It is hypothesized that BCG vaccination may help the selection of *M. tuberculosis* strains that resist BCG-induced immunity (Hermans et al., 1995; Parwati et al., 2010).

In this study among the 33 Pre-XDR TB cases, newly diagnosed Pre-XDR-TB cases were 10 (30.30%) in number. There are 8(24.24%) Cat 2 failure cases and 7 (21.21%) Cat 2 relapse cases 4 case (12.12%) of Cat 1 failure, 3 case (9.10%) of Cat1 relapse and only one case (3.03%) of Cat 1 non converter was also found. According to findings of the present study, Beijing was the major lineage to cause Category 2 treatment failure 4(12.12%) and Category 2 relapse 4(12.12%) rather than others.

Table 3

Phenotypic drug resistance according to mutations in the target gene and development of corresponding wild type and mutation bands among Pre-XDR-TB isolates (n = 33).

Phenotypic drug resistance	Wild type band		Developing mutation band	Mutation	No. (%) of mutations
	Developed	Not developed			
Fluoroquinolones (FQL)	gyr A WT 1*		not developed	not developed	0(0)
		gyr A	MUT 1**	A90V(GCG-GTG)	11(33.33)
		WT 2**	MUT 2**	S91P(TCG-CCG)	1(3.03)
		gyr A	MUT 3A	D94A(GAC-GCC)	0(0)
		WT 3**	MUT 3B**	D94N (GAC-AAC)/D94Y (GAC-TAC)	3(9.10)
			MUT 3C**	D94G(GAC-GGC)	13(39.39)
			MUT 3D	D94H(GAC-CAC)	0(0)
		gyr B WT**	MUT 1	N538D (AAC-GAC)	0(0)
			MUT 2**	E540V (GAA-GTA)	2(6.06)
Total Fluoroquinolones resistance					30(90.90)
Injectable 2nd line drug(ISL)	rrs WT 1		MUT 1 not developed	A1401G/C1402T	0(0)
	rrs WT 2***		MUT 2***	G1484T	3(9.10)
	eis WT 1		not developed	G-37T	0(0)
	eis WT 2		MUT1 not developed	C-14T	0(0)
			not developed	C-12T	0(0)
			not developed	G-10A	0(0)
	eis WT 3		not developed	C-2A	0(0)
Total Injectable 2nd line Drug resistance					3(9.10)
Total					33(100)

Note:

A = Alanine; V=Valine; S=Serine; D = Aspartic acid; N = Asparagine; Y = Tyrosine;G = Glycine; H = Histidine; E = Glutamic acid. T = Threonine, C = Cystine.

* The strain regarded as sensitive when all the wild type of bands were present and all the mutation type bands were absent.

** The presence of any mutation band regarded as resistance to the respective antibiotic. Absence of at least one of the wild type was also regarded as resistant.

*** The presence of both mutation band and corresponding wild type band regarded as heteroresistant strain or presence of more than one *M. tuberculosis* complex strain in specimen. Phenotypic Drug resistance according to mutations in the target gene and the corresponding wild.

Out of 16 Beijing strains, 6 (18.18%) were found in treatment failure cases (data not shown in table). There is also predominance of Beijing strains (50%) among new Pre-XDR TB cases. Previous data found that in Bangladesh MDR-TB prevalence is estimated to be 15 times higher in re-treatment cases than in new tuberculosis cases (Kendall et al., 2015). A study conducted in Bangladesh also found that Beijing strains were prevalent in treatment failure cases (Banu et al., 2004). Another study at western Maharashtra in India identified most of the strains as CAS but Beijing strains were found to be significantly associated with treatment failure (Chatterjee et al., 2010). In Vietnam, The Beijing genotype was significantly associated with treatment failure and relapse (Lan et al., 2003). Beijing genotype found as an independent risk factor for tuberculosis treatment failure in Indonesia (Parwati et al., 2010).

A clonal complex in this analysis was defined as a group of strains with no more than a single locus variation. On the other hand, Strains not found in the clonal complexes regarded as Singletons. A cluster was defined as two or more isolates from different patients with identical MIRU patterns, whereas non-clustered patterns were referred to as unique. Minimum spanning tree analysis through MIRU-VNTR plus web tool showed 4 clonal complexes. Largest clonal complex is formed by 5 Beijing strains. Another 3 clonal complex formed by Beijing and Delhi/CAS strains consisting of 2 strains of each lineage. Rest of the 22 isolates was remaining as Singletons. Four strains arranged in 2 cluster and these clustered isolates belongs to Beijing lineage.

Based on this finding, recent transmission index (RTI) was found in this study to 0.121 or 12.1%. The recent transmission index calculated through 'n methods' (Glynn et al., 1999). This index shows that, isolates in cluster is considered as a recent transmission and non-clustered isolates considered as reactivation. This index is variable between 0 (the least recent transmission) to 1 (maximum genotypes transmission). Low transmission rate (6.5%) was also found in another study held in rural area of Matlab, Bangladesh (Banu et al., 2012b). Drug resistance tuberculosis occurs via two mechanisms: acquisition and direct transmission (Caminero, 2008).

Recent data signify that a large number of cases were found to be due to direct transmission rather than acquisition (Nodieva et al., 2010). A recent study was held by incorporating data from 6 countries of different

geographic location with different TB burden ratio. The findings were that 96% of tuberculosis cases were due to direct transmission. The highest transmission (99%) was estimated in Uzbekistan and lowest (48%) was in Bangladesh though it is a country of high MDR-TB burden. The other countries were included in this study were Ethiopia, Malawi, Peru and Philippines showed the transmission rate 92%, 82%, 95% and 76% respectively (Kendall et al., 2015). In Iraq, a study found 33.6% transmission rate (Mustafa Ali et al., 2014). Low index found in the present and previous studies of Bangladesh indicates that the drug resistance tuberculosis is occurring in our area is by reactivation or acquisition rather than transmission.

The allelic diversity of each MIRU-VNTR locus in this study was evaluated by Hunter Gaston discriminatory index (HGDI). The lowest (0.036) discriminatory locus in this study was 154/MIRU02. The overall discriminatory index of the 24 locus MIRU-VNTR was 0.84.

EAI lineage is considered as ancestral lineage regarding presence of the *M. tuberculosis* specific deletion region (TbD1) region in their genome (Smith et al., 2009). Though TbD1 detection required deletion analysis, a correlation was found with the presence of ≥ 2 copy number of repeats on MIRU 24 locus with the presence of TbD1 (Gutierrez et al., 2006). So, MIRU 24 itself may be an explanatory locus to detect ancestral and modern lineages. In the present study two EAI strains belongs to ancestral lineage as they showed ≥ 2 copy number of repeats on MIRU24 locus. Gutierrez et al. also found the same findings in MIRU 24 loci (Gutierrez et al., 2006). A study in Singapore also found the similar findings (Sun et al., 2004). A study in London also found ≥ 2 copy number of repeats on MIRU 24 locus (Brown et al., 2010). The other lineages in our study showed only 1 copy of repeats at MIRU 24 and they correspond with the modern lineages.

Thirty-three Pre-XDR-TB isolates were identified by 2nd line LPA among them 90.90% were Fluoroquinolones resistant. Only 9.10% isolates showed resistant to injectable 2nd line drugs (ISL) (Amikacin, kanamycin and capreomycin). In a study conducted by Hu et al. in china, observed similar results that were higher number of Fluoroquinolones (FLQ) resistant pre-XDR-TB (61.53%) cases and ISL resistant Pre-XDR-TB cases were 38.46% (Hu et al., 2013). The higher number of FLQ resistant Tuberculosis may be the result of indiscriminate use of FLQs in human

Table 4
Correlation of Fluoroquinolones and injectable 2nd line drug resistance with MTB lineage among Pre-XDR-TB isolates (n = 33).

Lineage	Total No. (%)	Fluoroquinolones resistance										Injectable 2nd line drug resistance					
		gyr A					gyr B					rrs			eis		
		WT 1 Band absent	WT 2 Band absent, MUT1 band present**** No. (%)	WT 3 Band absent, MUT2 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 3 Band absent, MUT3 band present** No. (%)	WT 1 band absent, MUT1 band present No. (%)	WT 2 band absent, MUT2 band present No. (%)	WT 2 band absent, MUT2 band present No. (%)	WT 2 band absent, MUT2 band present No. (%)	WT band absent, MUT1 band present No. (%)	WT band absent, MUT1 band present No. (%)
Beijing	16 (48.48)	0(0)	5(15.15)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	2(6.06)	0(0)	0(0)
Delhi/CAS	5 (15.15)	0(0)	3(9.10)	0(0)	1(3.03)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	1(3.03)	0(0)	0(0)
LAM	4 (12.12)	0(0)	1(3.03)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)
EAI	3(9.10)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)
Haarlem	2(6.06)	0(0)	0(0)	1(3.03)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)
Cameroon	2(6.06)	0(0)	2(6.06)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)
NEW-1	1(3.03)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)
Total	33 (100)	0(0)	11(33.33)	1(3.03)	3(9.10)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	0(0)	3(9.10)	0(0)	0(0)

** The strain regarded as sensitive when all the wild type of bands were present and all the mutation type bands were absent.

** The presence of any mutation band regarded as resistance to the respective antibiotic. Absence of at least one of the wild type was also regarded as resistant (Hain Life science GmbH, Nehren, Germany)

*** The presence of both mutation band and corresponding wild type band regarded as heteroresistant strain or presence of more than one *M. tuberculosis* complex strain in specimen.

infections other than Tuberculosis (Adwani et al., 2016).

The present study showed that among Pre-XDR-TB isolates the most frequent single mutation detected was the substitution from adenine to guanine in gyrA at codon 94, (MUT3C) which led to an amino acid change from aspartic acid to glycine (39.39%). Another 9.10% Pre-XDR-TB isolates showed point mutation at codon 94 (MUT 3B) which result in amino acid change from aspartic acid to Arginine or Tyrosine. Both mutations suggest resistance of levofloxacin as well as Mutation associated with high-level increase in MIC (minimum inhibitory concentration) for.

Moxifloxacin. So that even at a high dose moxifloxacin as well as other Fluoroquinolones cannot be considered as an effective medicine for tuberculosis.

Another 33.33% Pre-XDR-TB isolates showed point mutation resulting substitution of cytosine to Thymine at codon 90 (MUT 1) which leads to an amino acid change from alanine to valine. Only 3.03% Pre-XDR-TB isolates showed point mutation at codon 91(MUT 2) result in amino acid change from Serine to Proline. Furthermore, in gyrB, only 6.06% FQ-resistant isolates had a change from adenine to thymine at codon 540 resulting amino acid change from glutamic acid to valine. These mutations suggest resistance of levofloxacin as well as mutation associated with at least low-level increase in MIC for moxifloxacin. So, moxifloxacin could be used at higher dose in treatment of tuberculosis. (Aubry et al., 2006).

Regarding resistance to second-line injectable drugs, all of the resistant isolates (9.10%) showed point mutation at position 1484 of the rrs gene resulting substitution of guanine to thymine that suggest cross-resistance to kanamycin, amikacin and capreomycin (Maus et al., 2005). Additionally, the presence of rrs WT1 and rrs WT2 alongside of rrs MUT2 (G1484T) suggest the tested specimen contains a heteroresistant strain or the tested specimen contains more than one *M. tuberculosis* complex strain may be due to mixed infection of the patient. (Lu et al., 2016) In a study conducted by Hu et al. in china, observed similar results, where most common mutation was the change from adenine to guanine at position 1401 of the rrs gene (Hu et al., 2013).

The current study detected higher number of FLQ resistant Pre-XDR-TB cases than ISL resistant Pre-XDR-TB. Data suggest that the incidence of FLQ resistant Pre-XDR-TB is increasing worldwide (Singhal et al., 2016). The higher number of FLQ resistant Pre -XDR-TB cases may be due to indiscriminate use of FLQs in most of the common infection including pneumonia and pyrexia of unknown origin, other than *M. tuberculosis* in our country. When FLQs are used as antibiotics they have two detrimental effects, first they have anti mycobacterial action which can delay the diagnosis of Tuberculosis. Secondly, when previously used as antibiotics they can lead to selection of FLQ resistant *M. tuberculosis* mutants (Adwani et al., 2016). The exposure to FLQ antibiotics are more common than ISLs as FLQs antibiotics are oral drugs which are readily available in open market without prescription. ISLs are aminoglycosides (amikacin, kanamycin and capreomycin) which are injectable antibiotics but also available without prescription in our country and are used for the treatment of bacterial infection other than TB. It may have contributed to the evolution of ISL resistant Pre-XDR-TB cases (Daniel et al., 2013).

Goal of this study was to identify the circulating MTB strains of different lineages among Pre-XDR pulmonary TB isolates. NTRL is the only national reference laboratory for tuberculosis in Bangladesh. Though study sample was not sufficient to give the actual situation of the circulating strains of whole country. But it will give a snapshot of the present situation. Beijing was the predominant lineage followed by Delhi/CAS in this study. Beijing was most predominant when drug resistance and urban population were the parameters. We found 2 Cameroon strains which was a specialist sub lineage and adapted in some endemic regions. Both of the patients harboring this strain remained for many years in those regions. Though the symptoms arouse after few years of returning, tuberculosis infection could be latent. To identify the source of the infection regular screening program should be

Table 5

Distribution of lineages among New and previous anti-TB treatment outcome categories among Pre- XDR –TB isolates (n = 33).

TB category	Total No. (%)	Lineage						
		Beijing No. (%)	Delhi/CAS No. (%)	LAM No. (%)	EAI No.(%)	Haarlem No.(%)	Cameroon No. (%)	NEW-1 No. (%)
New	10(30.30)	5(15.15)	1(3.03)	1(3.03)	0(0)	0(0)	2(6.06)	1(3.03)
Cat 1 Relapse	3(9.10)	1(3.03)	0(0)	1(3.03)	1(3.03)	0(0)	0(0)	0(0)
Cat 1 failure	4(12.12)	2(6.06)	1(3.03)	0(0)	1(3.03)	0(0)	0(0)	0(0)
Cat 2 Relapse	7(21.21)	4(12.12)	1(3.03)	1(3.03)	0(0)	1(3.03)	0(0)	0(0)
Cat 1 Non Converter	1(3.03)	0(0)	0(0)	0(0)	1(3.03)	0(0)	0(0)	0(0)
Cat 2 Failure	8(24.24)	4(12.12)	2(6.1)	1(3.03)	0(0)	1(3.03)	0(0)	0(0)
Total	33(100)	16(48.48)	5(15.15)	4(12.12)	3(9.10)	2(6.06)	2(6.06)	1(3.03)

launched when people come or return from other country.

Clustering of the isolates and clonal complexes found by minimum spanning tree indicates that the recent transmission was occurred only by Beijing strains and most of them were circulating in young age group. Though the low transmission rate indicates that national tuberculosis control program is on the right path to prevent transmission, Beijing strains were capable of escapes these measures. Beijing strains were also related with most of the treatment failure cases. So, large scale study should be launched to address the further different measures to control this hyper virulent Beijing lineage. Low transmission rate also indicates that there was a profound relationship between drug resistance and previous anti-TB treatment as most of the drug resistance TB is occurring through acquisition. The current study also found the previous anti-TB treatment as the significant risk factor. In this study 2nd Line LPA showed all of the Pre-XDR-TB isolates are resistant to Levofloxacin. In depth large scale nationwide molecular epidemiological study alongside of drug resistance survey is required to identify the factors for the acquisition of drug resistance and helps in modification of treatment strategy regarding Pre-XDR-TB treatment in a view to eliminate this notorious disease in our country.

Disclosure statement

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Submission declaration and verification

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The authors have seen and approved the final version of the manuscript being submitted. The manuscript is the author's original work, hasn't received prior publication and isn't under consideration for publication elsewhere.

Conflicts of interest

The authors have no co interests.

CRediT authorship contribution statement

Bayzid Bin Monir: Methodology, Validation, Data curation, Investigation, Writing – original draft, Software. **Sabia Shahin Sultana:** Data curation, Investigation, Writing – original draft, Software. **Shirin Tarafder:** Conceptualization, Methodology, Validation, Supervision, Writing – review & editing, Investigation.

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Appendix A. Supplementary data

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